Name_____ Date_____ Period_____

LESSON 5-3 · HOMEWORK PACKET · SHEET 1 OF 2

Trapezoids by Decomposing

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a trapezoid by decomposing it into a rectangle and triangles.

REMEMBER

- **Decompose** a trapezoid into a rectangle and one or two triangles.
- Find the area of each piece, then **add** the areas.
- When you cut off a rectangle, the triangle bases together equal the **difference** of the two parallel sides.

WORKED EXAMPLE

An isosceles trapezoid has parallel sides 6 m and 12 m and height 5 m. Find the area.

Step 1 Rectangle: 6 × 5 = 30

Step 2 Triangle bases: (12 − 6) ÷ 2 = 3 m each

Step 3 Triangles: 2 × (½ × 3 × 5) = 15

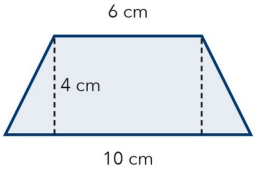
Step 4 Add: 30 + 15 = 45

Answer **45 m²**

PRACTICE

show your work · label units

1. The trapezoid is cut into a rectangle and two matching triangles.

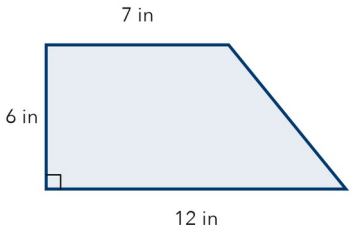


a. Area of the rectangle _____ cm²

b. Area of one triangle _____ cm²

Area of trapezoid_____ cm²

2. Decompose the trapezoid into a rectangle and one triangle. Find its area.



Show your work.

Answer_____ in²

3. Skate Ramp. The side of a skate ramp is a trapezoid with one square corner. Its parallel sides are 3 ft and 8 ft, and its height is 4 ft. Find the area by decomposing.

Show your work.

Answer_____ ft²

4. Table Top. A table top is a trapezoid with parallel sides of 5 ft and 11 ft and a height of 3 ft. It is cut into a rectangle and two matching triangles. Find its area.

Answer_____ ft²

5. Find the Error. A trapezoid has parallel sides 9 cm and 15 cm and height 4 cm. Tomás says each of its 2 matching triangles has a base of 6 cm. Explain his mistake and find the area.

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a trapezoid shape at home or outside (a lampshade side, a bag, a roof). Show someone how to cut it into a rectangle and triangles.

Name_____ Date_____ Period_____

LESSON 5-3 · HOMEWORK PACKET · SHEET 2 OF 2

The Trapezoid Area Formula

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a trapezoid using the trapezoid area formula.

REMEMBER

- **A = ½ × (b1 + b2) × h**. The bases b1 and b2 are the **parallel** sides.
- Add the bases first, then multiply by the height *h*, then take half.

WORKED EXAMPLE

Parallel sides 7 cm and 11 cm, height 6 cm. Find the area.

Step 1 Add the bases: 7 + 11 = 18

Step 2 Multiply by the height: 18 × 6 = 108

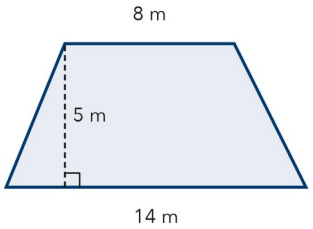
Step 3 Take half: 108 ÷ 2 = 54

Answer **54 cm²**

PRACTICE

show your work · label units

1. Find the area of the trapezoid.



Show your work.

Answer_____ m²

2. Find the area of each trapezoid with bases b1 and b2 and height *h*.

a. 4 ft, 10 ft; h = 6 ft _____ ft²

b. 5 in, 9 in; h = 8 in _____ in²

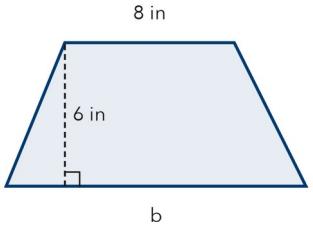
c. 6 m, 12 m; h = 3 m _____ m²

d. 3 cm, 7 cm; h = 9 cm _____ cm²

3. **Garden Plot.** A garden plot is a trapezoid with parallel sides of 16 ft and 24 ft and a height of 10 ft. One bag of mulch covers 25 ft². How many bags are needed?

Answer_____ bags

4. **Missing Base.** The area of the trapezoid is 60 in². Find the missing base *b*.



Show your work.

Answer_____ in

TEST PRACTICE

choose the best answer

5. A trapezoid has parallel sides of 6 cm and 10 cm and a height of 5 cm. What is its area?

A 21 cm² **B** 40 cm² **C** 80 cm² **D** 300 cm²

6. Which expression finds the area of a trapezoid with bases 3 m and 9 m and height 4 m?

A ½ × 3 × 9 × 4 **B** (3 + 9) × 4 **C** ½ × (3 + 9) × 4 **D** ½ × 3 + 9 × 4

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Measure a trapezoid shape at home, like the side of a bag. Estimate its area with the formula.